

AI chatbots mark the rise of generative artificial intelligence (AI): these tools have seen the fastest worker take-up of any new technology (Bick, Blandin and Deming, 2025), and controlled experiments demonstrate large productivity gains for users (Brynjolfsson, Li and Raymond, 2025; Noy and Zhang, 2023). Yet even as generative AI tools have become widespread, substantial disagreement remains about their labor market effects, with forecasts ranging from dramatic displacement within a few years (Amodei, 2025) to modest impacts over a decade (Acemoglu, 2025).¹ This disagreement partly reflects how little evidence exists on how labor markets actually absorb new technologies in their early stages. How does work reorganize at the onset of major technological change?

This paper argues that early labor market transformation under generative AI is best understood as *rapid currents under still waters*: substantial reorganization of work occurs as workplaces integrate AI chatbots and workers take on new AI-related tasks, absorbing change before it surfaces in earnings or recorded hours.

To study this process, we develop a unique data infrastructure linking workplace adoption and work reorganization to administrative labor market outcomes. In collaboration with Statistics Denmark, we conducted a series of large-scale, representative surveys of AI chatbot adoption in 11 highly exposed occupations—where labor market effects are likely to emerge earliest and strongest—linking responses to administrative labor market records.² Our latest survey round, conducted in late 2024, includes responses from 25,000 workers across 7,000 workplaces. These linked survey–register data allow us to examine both recorded labor market outcomes—such as earnings and hours—and adjustments that typically go unrecorded, including changes in job tasks and work organization. Crucially, even within these highly exposed occupations, many otherwise similar workplaces and workers have taken markedly different adoption paths, allowing us to examine how chatbot

¹Anthropic CEO Dario Amodei predicts that AI could eliminate half of all entry-level white-collar jobs, driving unemployment to 10–20% within the next one to five years (Amodei, 2025). In contrast, Acemoglu (2025) predicts that AI will increase annual TFP growth by less than 0.07 percentage points over the next ten years.

²Our list of occupations is accountants, customer support specialists, financial advisors, HR professionals, IT support specialists, journalists, legal professionals, marketing professionals, office clerks, software developers, and teachers.

on monthly earnings, hours worked, and occupations through December 2024—two years after the launch of ChatGPT. Using a difference-in-differences framework, we compare adopters and non-adopters before and after the arrival of AI chatbots, estimating impacts at both the worker and workplace levels.

We first analyze workers who use AI chatbots. If chatbot use enhances individual productivity, a natural question is whether and how such gains affect their earnings (Krueger, 1993). The answer is theoretically ambiguous, depending on whether adoption is worker- or firm-driven (Becker, 1964), whether it displaces users from higher-value tasks (Autor and Thompson, 2025), whether chatbot use is valued as a nonpecuniary amenity (Rosen, 1986), and on the structure of wage setting (Card et al., 2018).

We find precisely null effects of AI chatbot use on earnings and hours. Difference-in-differences estimates for earnings, recorded hours, and wages all center on zero, with confidence intervals ruling out average effects larger than 2%. These null results hold even among workers who use chatbots daily, report substantial productivity gains, perform new AI-related tasks, or work in firms that encourage and invest in chatbots. Moreover, we find no significant effects in any of our 11 occupations, including those with flexible, decentralized wage-setting.

While chatbots have not led to earnings gains for users on average, we find a clear link to occupational mobility: since the arrival of AI chatbots, adopters are significantly more likely to have switched occupations, working about 4% of a full-time equivalent more in their most recent occupation by December 2024. These transitions are most pronounced into IT support specialist and clerical (office support) roles—occupations in which workers have greater discretion over their toolsets and that do not require formal educational credentials. These patterns align with experimental evidence that chatbots are especially beneficial for workers with less prior expertise (Brynjolfsson, Li and Raymond, 2025; Noy and Zhang, 2023) and support the view that generative AI helps workers move into new occupations by granting access to otherwise scarce expertise (Autor, 2024). In particular, adopters who switch occupations tend to move into roles with higher wage premia and